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CHEM-STUD AND POWER-FAST ADHESIVE ANCHOR SYSTEMS

POWERS FASTENERS, INC.
TWO POWERS SQUARE
NEW ROCHELLE, NEW YORK 10801

1.0 SUBJECT

Chem-Stud and Power-Fast Adhesive Anchor Systems.

2.0 DESCRIPTION

2.1 Power-Fast:

2.1.1 General: The Power-Fast Adhesive Anchor System consists of a two-component structural epoxy that is used to bond threaded rods, fully threaded bolts, or reinforcing steel in predrilled holes in normal-weight, stone-aggregate concrete. The Power-Fast Adhesive is available in a Standard Set and a Fast Set which has an additive admixture for quicker curing time as noted in Table 2.

2.1.2 Materials:

2.1.2.1 Adhesive: Power-Fast adhesive is a two-component epoxy that is packaged in equal-volume side-by-side plastic cartridges or pumps. The components include a base resin and hardener that are mixed in a one-to-one ratio as they are dispensed through a disposable static mixing nozzle (supplied by Powers Fasteners, Inc.) or manually through a pump cartridge. Fast Set and Standard Set versions are available. The minimum shelf life of the adhesive is two years when stored in a dry environment at temperatures between 40°F to 90°F (4.4°C to 32.2°C).

2.1.2.2 Anchor Rods: Threaded rods range in diameter from $\frac{3}{8}$ to $1\frac{1}{4}$ inches (9.5 to 31.8 mm) and are manufactured from carbon steel in compliance with ASTM A 36, ASTM A 307, or ASTM A 193 Grade B7. Stainless steel threaded rod is manufactured from AISI Type 304 or 316 material. Fully threaded bolts (Chem-Bolt) range in diameter from $\frac{3}{8}$ to $\frac{3}{4}$ inch (9.5 to 19.1 mm) and are manufactured from carbon steel in compliance with SAE Grade 2 or Grade 5. Fully threaded stainless steel bolts are manufactured from AISI Type 304 material. Specification and installation requirements for threaded rod are listed in Table 1.

2.1.2.3 Reinforcing Bars: Deformed reinforcing bars range in size from No. 3 to No. 10 and are manufactured from steel conforming to ASTM A 615, A 616, A 617, or A 706, Grade 40 or 60. Specification and installation requirements for reinforcing bars are listed in Table 8.

2.1.2.4 Concrete: Anchors are installed in normal-weight, stone aggregate concrete having a compressive strength, at the time of anchor installation, that is noted in Tables 4, 5, 6, 7, 10, and 11.

2.2 Chem-Stud:

2.2.1 General: The Chem-Stud Anchor System consists of a two-part resin capsule that is used to bond chisel-pointed threaded rods in predrilled holes in normal-weight, stone-aggregate concrete.

2.2.2 Materials:

2.2.2.1 Adhesive: The Chem-Stud capsule contains pre-measured amounts of resin and hardener in two sealed glass tubes. The outer glass tube is filled with vinylester resin and quartz aggregate. A smaller sealed glass tube is suspended in the resin and is filled with a hardening catalyst. The adhesive components are mixed by spinning a chisel-pointed anchor rod into the capsule, using a rotary hammer. A specific capsule size is provided for each diameter of anchor rod. The shelf life of the capsules is two years when stored in a dry environment at temperatures under 71°F (21.1°C).

2.2.2.2 Anchor Rods: Chem-Stud threaded anchor rods have a chisel point for proper mixing of the adhesive. Threaded rods range in diameter from $\frac{3}{8}$ to $1\frac{1}{4}$ inches (9.5 to 31.8 mm), and are manufactured from carbon steel in compliance with ASTM A 36, ASTM A 307, or ASTM A 193 Grade B7. Stainless steel threaded rod is manufactured from AISI Type 304 or 316 material. Specification and installation requirements for threaded rod are listed in Table 12.

2.2.2.3 Concrete: Anchors are installed in normal-weight, stone-aggregate concrete having a compressive strength that is noted in Tables 15 and 16.

2.3 Design:

Allowable static loads for anchors installed in accordance with this report are listed in Tables 4, 5, 6, 7, 10, 11, 15 and 16. The allowable tension and shear loads for the adhesives must be adjusted for elevated temperatures in accordance with Figures 1 and 2, and adjusted for spacing and edge effects in accordance with Tables 3, 9 and 14. Allowable loads for anchors subjected to combined shear and tension forces are determined by the following equation:

$$(P_s/P_t) + (V_s/V_t) \leq 1$$

where:

P_s = Applied service tension load.

P_t = Allowable service tension load.

V_s = Applied service shear load.

V_t = Allowable service shear load.

Anchors are not permitted to be subjected to vibratory or shock loads, such as those encountered by supports for reciprocating engines, crane loads and moving loads due to vehicles.

Anchors are permitted for dead load, live load, seismic load and wind load applications, as noted in the tables. The allow-

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able load values for the Power-Fast Epoxy Adhesive, Standard Set, installed with reinforcing steel may be increased for short-term loads such as wind or earthquake loads. The allowable load values for Power-Fast Epoxy Adhesive, Fast Set, used with reinforcing steel is permitted to resist short-term wind loads only.

2.4 Installation:

2.4.1 General: Anchors must be installed in accordance with the manufacturer's published installation instructions and the requirements of this report. Anchor holes are predrilled into normal-weight concrete at a predetermined depth, using hand-held rotary hammer drills with a carbide-tipped drill bit that complies with ANSI B 212.15-1994. The anchor hole must be cleaned of dust and debris using compressed air and a nylon brush. Holes are permitted to be dry or damp, but all standing water must be removed. The hole diameter, embedment depth, spacing, edge distance, and concrete must comply with the requirements in Tables 1 through 16.

2.4.2 Power-Fast: The Power-Fast epoxy is injected into the hole from a static mixing nozzle, starting at the bottom, until the hole is approximately one-half full. The threaded rod, fully threaded bolt, or reinforcing bar is inserted into the hole, while rotating, to the required embedment depth. The Power-Fast epoxy must be allowed to cure in accordance with Table 2 prior to the application of the design load.

2.4.3 Chem-Stud: The adhesive capsule is inserted completely into the anchor hole. Using a rotary hammer, the chisel-pointed threaded rod is spun into the adhesive capsule to the required embedment depth. The Chem-Stud adhesive must be allowed to cure in accordance with Table 13 prior to the application of the design load.

2.4.4 Special Considerations: The anchors may be used within fire-resistive construction, provided the anchors only resist wind or seismic forces. The anchors cannot be used to resist tension loads in ceiling or wall installations unless special consideration is given to fire-exposure conditions.

2.5 Special Inspection:

Adhesive anchor installations require special inspection in accordance with Section 1701 of the UBC. The special inspector must record the product description, including adhesive; expiration date; concrete type and strength; drill bit dimensions in compliance with this report; hole diameter, depth and cleanliness; anchor rod diameter, length and steel grade; adhesive application; anchor embedment; and verification that the anchor installation is in accordance with the manufacturer's published installation instructions and this report.

2.6 Identification:

The adhesive anchors are identified by packaging bearing the manufacturer's name (Powers Fasteners), the product name, the material type, the batch number, the expiration date, and the evaluation report number (ICBO ES ER-4514).

3.0 EVIDENCE SUBMITTED

Data complying with the ICBO ES Acceptance Criteria for Adhesive Anchors in Concrete and Masonry Elements (AC58), dated January 1999, including creep, in-service temperature, dampness, freezing and thawing, and seismic data.

4.0 FINDINGS

That the Chem-Stud and Power-Fast Adhesive Anchor Systems described in this report comply with the 1997 Uniform Building Code™, subject to the following conditions:

- 4.1 The anchors are installed in accordance with the manufacturer's installation instructions and this report.
- 4.2 Anchors are installed in holes predrilled into concrete using a carbide-tipped drill bit manufactured within the range of maximum and minimum drill tip dimensions of ANSI B212.15-1994 for the allowable values set forth in this report.
- 4.3 Calculations and details showing compliance with this report are submitted to the building official for approval.
- 4.4 Special inspection in accordance with Section 2.5 of this report is provided for all anchor installations.
- 4.5 Anchors in concrete are limited to installation in uncracked concrete [which is defined as concrete subjected to maximum 170-psi (1172 kPa) tensile stress induced by external loads or deformations].
- 4.6 Anchors are not subjected to vibratory or shock loads, such as those encountered by supports for reciprocating engines or crane loads.
- 4.7 Anchors are not used in conjunction with fire-resistive construction, except as noted in Section 2.2.4.
- 4.8 Anchors are not used to resist tension forces in ceilings or wall installations, unless special consideration is given to fire-exposure conditions.
- 4.9 Maximum allowable shear and tension loads are as set forth in this report. Allowable loads must be adjusted for temperature in accordance with Figure 1 or 2.
- 4.10 Use of Power-Fast Epoxy Adhesive and Chem-Stud Adhesive anchors in concrete to resist earthquake loads, wind loads, dead loads or live loads is permitted as noted in the tables. The tabulated allowable load values in the tables may be increased by $33\frac{1}{3}$ percent for short-term loads, such as wind or earthquake loads, in accordance with Section 1612.3.3 of the code, as permitted in the tables. The allowable load values for the Power-Fast Epoxy Adhesive, Standard Set, installed with threaded rod, fully threaded bolts, or reinforcing steel may be increased for short-term loads, such as wind or earthquake loads. The allowable load values for the Power-Fast Epoxy Adhesive, Fast Set, installed with threaded rod or fully threaded bolts is permitted for short-term loads, such as those resulting from wind or earthquake forces only. The allowable load values for Power-Fast Epoxy Adhesive, Fast Set, used with reinforcing steel is permitted for short-term wind loads only.
- 4.11 Anchors are limited to interior use, except that installation, as noted in the tables, in severe, moderate, or negligible exterior weathering locations, in accordance with Figure 21-1-1 of UBC Standard 21-1, is permitted when stainless steel threaded rods are installed.
- 4.12 Anchors are permitted in dry-hole or damp-hole installations.
- 4.13 Anchors are manufactured at Two Powers Square, New Rochelle, New York, with quality control inspections performed by CEL Consulting (AA-639).

This report is subject to re-examination in one year.

TABLE 1—SPECIFICATIONS AND DETAILS FOR INSTALLATION OF THREADED ROD AND THREADED BOLTS IN CONCRETE USING POWER-FAST EPOXY ADHESIVE

PROPERTY	ROD DIAMETER						
	3/8 inch	1/2 inch	5/8 inch	3/4 inch	7/8 inch	1 inch	1 1/4 inches
A_b [Nominal area of rod (in. ²)]	0.1105	0.1963	0.3068	0.4418	0.6013	0.7854	1.2772
BD [Nominal bit diameter (in.)]	7/16	9/16	3/4	7/8	1	1 1/8	1 3/8
E [Minimum anchor embedment (in.)]	1 1/2	2	2 1/2	3	3 1/2	4	5
T [Maximum tightening torque (ft.-lbf.)]	20	40	90	160	200	300	600

For **SI**: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 foot-lbf. = 1.35 N-m.

TABLE 2—MANUFACTURER'S RECOMMENDED CURE TIME FOR POWER-FAST EPOXY ADHESIVE

BASE MATERIAL TEMPERATURE ¹ (°F)	MAXIMUM GEL TIME ² (minutes)		MINIMUM CURING TIME ³ (hours)		FULL CURING TIME ⁴ (hours)	
	Fast Set	Std. Set	Fast Set	Std. Set	Fast Set	Std. Set
40	NA	60	NA	16	NA	48
60	20	45	3	7	24	36
75	15	35	2	6	24	24
90	10	20	1 1/2	4	16	24

For **SI**: t°C = 5/9(t°F – 32).

NA = Not available.

¹This is the temperature as measured in the base material.

²The gel time is the maximum time during which the epoxy can be dispensed before it begins to set.

³Anchors must not be disturbed before the minimum curing time occurs. When the minimum cure time is achieved, the fixture can be positioned.

⁴The full curing time is the minimum time required for the epoxy to achieve its load capacities.

TABLE 3—REDUCTIONS FOR SPACING AND EDGE DISTANCES FOR THREADED ROD OR FULLY THREADED BOLTS INSTALLED IN CONCRETE WITH POWER-FAST EPOXY ADHESIVE

DIMENSION / LOAD TYPE	DISTANCE FOR FULL ANCHOR CAPACITY ¹	MINIMUM DISTANCE ²	REDUCTION FACTOR ³
SPACING (s) BETWEEN ANCHORS – TENSION AND SHEAR LOADS	$s_{cr} = 16D$	$s_{min} = 8D$	$F_t = F_v = 0.72$
EDGE DISTANCE (c) – TENSION LOADS	$c_{cr} = 10D$	$c_{min} = 4D$	$F_t = 0.56$
EDGE DISTANCE (c) – SHEAR LOADS	$c_{cr} = 12D$	$c_{min} = 4D$	$F_v = 0.21$

¹The listed values are the minimum distances required to obtain the load values listed in Tables 4, 5, 6, and 7. D = Anchor diameter. When adjacent anchors are different sizes or embedments, use largest value for D .

²The listed values are the minimum distances at which the anchor can be set, when load values are adjusted with the reduction factor noted in this table.

³Load values in the tables are multiplied by the reduction factor with the reduction factor noted in this table. Linear interpolation is allowed for edge distances and spacing that fall between critical and minimum distances. When a group of anchors is affected by both reduced spacing and reduced edge distance, the edge and spacing reduction factors must be combined (multiplied).

TABLE 4—ALLOWABLE TENSION LOADS FOR THREADED RODS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8}

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c=2,000$ psi	$f'_c=3,000$ psi	$f'_c=4,000$ psi	$f'_c=5,000$ psi	ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
3/8	7/16	6	3 3/4	1 1/2	865	980	1,110	1,300	2,115	2,185	4,555	3,630
				1 7/8	1,170	1,336	1,477	1,666				
				2 1/4	1,475	1,692	1,844	2,032				
				2 5/8	1,780	2,048	2,211	2,398				
				3	2,085	2,404	2,578	2,764				
				3 3/8	2,390	2,760	2,945	3,130				
				3 3/4	2,612	2,986	3,125	3,286				
				4 1/8	2,834	3,212	3,305	3,442				
				4 1/2	3,056	3,438	3,485	3,598				
				4 7/8	3,278	3,664	3,665	3,754				
1/2	9/16	8	5	5 1/4	3,500	3,890	3,845	3,910	3,755	3,885	8,095	6,470
				2	1,200	1,810	2,705	3,615				
				2 1/2	1,609	2,128	3,107	3,980				
				3	2,018	2,446	3,509	4,345				
				3 1/2	2,427	2,764	3,911	4,710				
				4	2,836	3,082	4,313	5,075				
				4 1/2	3,245	3,400	4,715	5,440				
				5	3,640	3,939	5,094	5,777				
				5 1/2	4,035	4,478	5,473	6,114				
				6	4,430	5,017	5,852	6,451				
				6 1/2	4,825	5,556	6,231	6,788				
				7	5,220	6,095	6,610	7,125				

(Continued)

**TABLE 4—ALLOWABLE TENSION LOADS FOR THREADED RODS INSTALLED
IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8}—(Continued)**

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, S_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c=2,000$ psi	$f'_c=3,000$ psi	$f'_c=4,000$ psi	$f'_c=5,000$ psi	ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
$5/8$	$3/4$	10	$6\frac{1}{4}$	$2\frac{1}{2}$	1,725	2,070	2,350	2,625	5,870	6,075	12,655	10,130
				$3\frac{1}{8}$	2,347	2,768	3,050	3,329				
				$3\frac{3}{4}$	2,969	3,466	3,750	4,033				
				$4\frac{3}{8}$	3,591	4,164	4,450	4,737				
				5	4,213	4,862	5,150	5,441				
				$5\frac{5}{8}$	4,835	5,560	5,850	6,145				
				$6\frac{1}{4}$	5,409	6,050	6,518	6,830				
				$6\frac{7}{8}$	5,983	6,540	7,186	7,515				
				$7\frac{1}{2}$	6,557	7,030	7,854	8,200				
				$8\frac{1}{8}$	7,131	7,520	8,522	8,885				
				$8\frac{3}{4}$	7,705	8,010	9,190	9,570				
$3/4$	$7/8$	12	$7\frac{1}{2}$	3	2,400	3,050	3,710	4,360	8,455	8,750	18,225	12,400
				$3\frac{3}{4}$	3,328	4,114	4,844	5,642				
				$4\frac{1}{2}$	4,256	5,178	5,978	6,924				
				$5\frac{1}{4}$	5,184	6,242	7,112	8,206				
				6	6,112	7,306	8,246	9,488				
				$6\frac{3}{4}$	7,040	8,370	9,380	10,770				
				$7\frac{1}{2}$	7,620	8,972	9,888	11,106				
				$8\frac{1}{4}$	8,200	9,574	10,396	11,442				
				9	8,780	10,176	10,904	11,778				
				$9\frac{3}{4}$	9,360	10,778	11,412	12,114				
				$10\frac{1}{2}$	9,940	11,380	11,920	12,450				
$7/8$	1	14	$8\frac{3}{4}$	$3\frac{1}{2}$	2,730	3,330	3,895	4,455	11,510	11,905	24,805	16,860
				$4\frac{3}{8}$	3,818	4,584	5,224	5,859				
				$5\frac{1}{4}$	4,906	5,838	6,553	7,263				
				$6\frac{1}{8}$	5,994	7,092	7,882	8,667				
				7	7,082	8,346	9,211	10,071				
				$7\frac{7}{8}$	8,170	9,600	10,540	11,475				
				$8\frac{3}{4}$	9,154	10,752	11,794	12,830				
				$9\frac{5}{8}$	10,138	11,904	13,048	14,185				
				$10\frac{1}{2}$	11,122	13,056	14,302	15,540				
				$11\frac{3}{8}$	12,106	14,208	15,556	16,895				
				$12\frac{1}{4}$	13,090	15,360	16,810	18,250				
1	$1\frac{1}{8}$	16	10	4	3,385	4,085	4,695	5,300	15,035	15,550	32,400	22,020
				5	4,696	5,694	6,593	7,486				
				6	6,007	7,303	8,491	9,672				
				7	7,318	8,912	10,389	11,858				
				8	8,629	10,521	12,287	14,044				
				9	9,940	12,130	14,185	16,230				
				10	10,897	13,142	15,078	17,005				
				11	11,854	14,154	15,971	17,780				
				12	12,811	15,166	16,864	18,555				
				13	13,768	16,178	17,757	19,330				
				14	14,725	17,190	18,650	20,105				
$1\frac{1}{4}$	$1\frac{3}{8}$	20	$12\frac{1}{2}$	5	4,500	6,020	7,690	9,695	23,485	24,295	50,615	34,420
				$6\frac{1}{4}$	6,246	8,156	10,348	12,801				
				$7\frac{1}{2}$	7,992	10,292	13,006	15,907				
				$8\frac{3}{4}$	9,738	12,428	15,664	19,013				
				10	11,484	14,564	18,322	22,119				
				$11\frac{1}{4}$	13,230	16,700	20,980	25,225				
				$12\frac{1}{2}$	15,158	18,822	22,948	27,042				
				$13\frac{3}{4}$	17,086	20,944	24,916	28,859				
				15	19,014	23,066	26,884	30,676				
				$16\frac{1}{4}$	20,942	25,188	28,852	32,493				
				$17\frac{1}{2}$	22,870	27,310	30,820	34,310				

(Continued)

Footnotes to Table 4

For **SI**: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength.

²For Earthquake and wind application with A 193 Grade B7 threaded rod, the lesser of the tabulated bond strength and the allowable steel strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. The threaded rod has a diameter/embedment ratio of 12, maximum. For earthquake and wind application with SS304/316 threaded rod, reduce the allowable steel strength using a load factor of 0.92. The lesser of this reduced steel strength and tabulated bond strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. For earthquake application with A 36 and A 307 threaded rod, reduce the allowable steel strength using a load factor of 0.48. The lesser of this reduced steel strength and tabulated bond strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. For wind application with A 36 and A 307 threaded rod, the allowable load must be the lesser of the tabulated bond strength and the allowable steel strength.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 3. The reduction factors listed in Table 3 must be applied to the values listed in the bond and concrete strength column.

⁴The tabulated shear and tension values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads for the Standard Set Power-Fast adhesive, based on bond strength, are calculated based on a safety factor of 4.0. When using the Fast Set Power-Fast adhesive, the bond strength must be reduced by 25 percent based on a safety factor of 5.33.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 1 for applicable load factors.

⁸Special inspection in accordance with Section 2.5 must be provided for all anchor installations.

**TABLE 5—ALLOWABLE SHEAR LOADS FOR THREADED RODS INSTALLED IN
NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8}**

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c=2,000$ psi Minimum	ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
$3/8$	$7/16$	6	$4\frac{1}{2}$	$1\frac{1}{2}$	1,150	1,090	1,125	2,345	1,870
				$1\frac{7}{8}$	1,220				
				$2\frac{1}{4}$	1,290				
				$2\frac{5}{8}$	1,360				
				3	1,430				
				$3\frac{3}{8}$	1,500				
$1/2$	$9/16$	8	6	2	1,700	1,940	2,000	4,170	3,330
				$2\frac{1}{2}$	2,000				
				3	2,300				
				$3\frac{1}{2}$	2,600				
				4	2,900				
				$4\frac{1}{2}$	3,200				
$5/8$	$3/4$	10	$7\frac{1}{2}$	$2\frac{1}{2}$	2,400	3,025	3,130	6,520	5,210
				$3\frac{1}{8}$	3,060				
				$3\frac{3}{4}$	3,720				
				$4\frac{3}{8}$	4,380				
				5	5,040				
				$5\frac{5}{8}$	5,700				
$3/4$	$7/8$	12	9	3	3,600	4,355	4,505	9,390	6,390
				$3\frac{3}{4}$	4,160				
				$4\frac{1}{2}$	4,720				
				$5\frac{1}{4}$	5,280				
				6	5,840				
				$6\frac{3}{4}$	6,400				
$7/8$	1	14	$10\frac{1}{2}$	$3\frac{1}{2}$	3,500	5,930	6,135	12,780	8,680
				$4\frac{3}{8}$	4,640				
				$5\frac{1}{4}$	5,780				
				$6\frac{1}{8}$	6,920				
				7	8,060				
				$7\frac{7}{8}$	9,200				
1	$1\frac{1}{8}$	16	12	4	4,600	7,745	8,010	16,690	11,340
				5	6,180				
				6	7,760				
				7	9,340				
				8	10,920				
				9	12,500				
$1\frac{1}{4}$	$1\frac{3}{8}$	20	15	5	5,500	12,100	12,515	26,075	17,730
				$6\frac{1}{4}$	8,000				
				$7\frac{1}{2}$	10,500				
				$8\frac{3}{4}$	13,000				
				10	15,500				
				$11\frac{1}{4}$	18,000				

(Continued)

Footnotes to Table 5

For **SI**: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength tabulated above.

²For short-term loading due to earthquakes or wind, allowable shear loads for anchors with A 36 and A 307 threaded rods may be increased by 33¹/₃%. A 36 and A 307 values must be used instead of listed values for higher strength steels for these loading conditions.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 3. The reduction factors listed in Table 3 must be applied to the values listed in the bond and concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads for the Standard Set Power-Fast adhesive, based on bond strength, are calculated based on a safety factor of 4.0. When using the Fast Set Power-Fast adhesive, the bond strength must be reduced by 25 percent based on a safety factor of 5.33.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 1 for applicable load factors.

⁸Special inspection in accordance with Section 2.5 must be provided for all anchor installations

TABLE 6—ALLOWABLE TENSION LOADS FOR FULLY THREADED BOLTS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8,9}

ANCHOR DIAMETER (inch)	BIT DIAMETER (inch)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c=2,000$ psi	$f'_c=3,000$ psi	$f'_c=4,000$ psi	$f'_c=5,000$ psi	GRADE 1	GRADE 2	GRADE 5	304 / 316 SS
$\frac{3}{8}$	$\frac{7}{16}$	6	$3\frac{3}{4}$	$1\frac{1}{2}$	865	980	1,110	1,300	2,180	2,685	4,355	3,630
				$1\frac{7}{8}$	1,170	1,336	1,477	1,666				
				$2\frac{1}{4}$	1,475	1,692	1,844	2,032				
				$2\frac{5}{8}$	1,780	2,048	2,211	2,398				
				3	2,085	2,404	2,578	2,764				
				$3\frac{3}{8}$	2,390	2,760	2,945	3,130				
				$3\frac{3}{4}$	2,612	2,986	3,125	3,286				
				$4\frac{1}{8}$	2,834	3,212	3,305	3,442				
				$4\frac{1}{2}$	3,056	3,438	3,485	3,598				
				$4\frac{7}{8}$	3,278	3,664	3,665	3,754				
$\frac{1}{2}$	$\frac{9}{16}$	8	5	$5\frac{1}{4}$	3,500	3,890	3,845	3,910	3,880	4,785	7,760	6,470
				2	1,200	1,810	2,705	3,615				
				$2\frac{1}{2}$	1,609	2,128	3,107	3,980				
				3	2,018	2,446	3,509	4,345				
				$3\frac{1}{2}$	2,427	2,764	3,911	4,710				
				4	2,836	3,082	4,313	5,075				
				$4\frac{1}{2}$	3,245	3,400	4,715	5,440				
				5	3,640	3,939	5,094	5,777				
				$5\frac{1}{2}$	4,035	4,478	5,473	6,114				
				6	4,430	5,017	5,852	6,451				
$\frac{5}{8}$	$\frac{3}{4}$	10	$6\frac{1}{4}$	$6\frac{1}{2}$	4,825	5,556	6,231	6,788	6,080	7,495	12,155	10,130
				7	5,220	6,095	6,610	7,125				
				$2\frac{1}{2}$	1,725	2,070	2,350	2,625				
				$3\frac{1}{8}$	2,347	2,768	3,050	3,329				
				$3\frac{3}{4}$	2,969	3,466	3,750	4,033				
				$4\frac{3}{8}$	3,591	4,164	4,450	4,737				
				5	4,213	4,862	5,150	5,441				
				$5\frac{5}{8}$	4,835	5,560	5,850	6,145				
				$6\frac{1}{4}$	5,409	6,050	6,518	6,830				
				$6\frac{7}{8}$	5,983	6,540	7,186	7,515				
$\frac{3}{4}$	$\frac{7}{8}$	12	$7\frac{1}{2}$	$7\frac{1}{2}$	6,557	7,030	7,854	8,200	8,790	10,795	17,500	12,400
				$8\frac{1}{8}$	7,131	7,520	8,522	8,885				
				$8\frac{3}{4}$	7,705	8,010	9,190	9,570				
				3	2,400	3,050	3,710	4,360				
				$3\frac{3}{4}$	3,328	4,114	4,844	5,642				
				$4\frac{1}{2}$	4,256	5,178	5,978	6,924				
				$5\frac{1}{4}$	5,184	6,242	7,112	8,206				
				6	6,112	7,306	8,246	9,488				
				$6\frac{3}{4}$	7,040	8,370	9,380	10,770				
				$7\frac{1}{2}$	7,620	8,972	9,888	11,106				
$\frac{3}{4}$	$\frac{7}{8}$	12	$7\frac{1}{2}$	$8\frac{1}{4}$	8,200	9,574	10,396	11,442				
				9	8,780	10,176	10,904	11,778				
				$9\frac{3}{4}$	9,360	10,778	11,412	12,114				

For **SI**: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength.

²For earthquake applications with Grades 1 and 2 threaded bolt, reduce the allowable steel strength using a load factor of 0.48. For earthquake applications with Grade 5 threaded bolts, reduce the allowable steel strength using a load factor of 0.84. For earthquake application with SS 304/316 threaded bolts, reduce the allowable steel strength using a load factor of 0.92. The lesser of these reduced steel strengths and tabulated bond strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 3. The reduction factors listed in Table 3 must be applied to the values listed in the bond or concrete strength column.

⁴The tabulated shear and tension values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads for the Standard Set Power-Fast adhesive, based on bond strength, are calculated based on a safety factor of 4.0. When using the Fast Set Power-Fast adhesive, the bond strength must be reduced by 25 percent based on a safety factor of 5.33.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷According to SAE, Grade 2 requirements for $3/8$ inch to $3/4$ inch diameters apply only to bolts 6 inches long and shorter. For bolts longer than 6 inches, Grade 1 steel requirements apply.

⁸Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 1 for applicable load factors.

⁹Special inspection in accordance with Section 2.5 must be provided for all anchor installations.

TABLE 7—ALLOWABLE SHEAR LOADS FOR FULLY TREADED BOLTS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8,9}

ANCHOR DIAMETER (inch)	BIT DIAMETER (inch)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c=2000$ psi Minimum	GRADE 1	GRADE 2	GRADE 5	304 / 316 SS
$3/8$	$7/16$	6	$4\frac{1}{2}$	$1\frac{1}{2}$	1,150	1,120	1,385	2,245	1,870
				$1\frac{7}{8}$	1,220				
				$2\frac{1}{4}$	1,290				
				$2\frac{5}{8}$	1,360				
				3	1,430				
				$3\frac{3}{8}$	1,500				
$1/2$	$9/16$	8	6	2	1,700	1,200	2,465	3,990	3,330
				$2\frac{1}{2}$	2,000				
				3	2,300				
				$3\frac{1}{2}$	2,600				
				4	2,900				
				$4\frac{1}{2}$	3,200				
$5/8$	$3/4$	10	$7\frac{1}{2}$	$2\frac{1}{2}$	2,400	3,130	3,860	6,260	5,210
				$3\frac{1}{8}$	3,060				
				$3\frac{3}{4}$	3,720				
				$4\frac{3}{8}$	4,380				
				5	5,040				
				$5\frac{5}{8}$	5,700				
$3/4$	$7/8$	12	9	3	3,600	4,510	5,560	9,015	6,390
				$3\frac{3}{4}$	4,160				
				$4\frac{1}{2}$	4,720				
				$5\frac{1}{4}$	5,280				
				6	5,840				
				$6\frac{3}{4}$	6,400				

For **SI**: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength.

²For short-term loading due to wind, allowable shear loads for Grade 1 or Grade 2 bolts may be increased by 33 $\frac{1}{3}$ %. Grade 1 or Grade 2 values must be used instead of listed values for higher strength steels for these loading conditions. For earthquake applications, the lesser of the bond strength and the allowable steel strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 3. The reduction factors listed in Table 3 must be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation.

⁵Allowable loads for the Standard Set Power-Fast adhesive, based on bond strength, are calculated based on a safety factor of 4.0. When using the Fast Set Power-Fast adhesive, the bond strength must be reduced by 25 percent based on a safety factor of 5.33.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷According to SAE, Grade 2 requirements for $3/8$ -inch- to $3/4$ -inch-diameter bolts apply only to bolts 6 inches long and shorter. For bolts longer than 6 inches, Grade 1 steel requirements apply.

⁸Adhesive anchors experience a reduction and shear capacity with increased ambient temperatures. See Figure 1 for applicable load factors.

⁹Special inspection in accordance with Section 2.5 must be provided for all anchor installations.

TABLE 8—SPECIFICATIONS AND DETAILS FOR INSTALLATION OF REINFORCING BAR IN CONCRETE USING POWER-FAST EPOXY ADHESIVE

PROPERTY	REINFORCING BAR SIZE							
	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10
A_{br} [Nominal area of rebar (in. ²)]	0.110	0.200	0.310	0.440	0.600	0.790	1.000	1.270
BD [Nominal bit diameter (in.)]	$\frac{1}{2}$	$\frac{5}{8}$	$\frac{3}{4}$	$\frac{7}{8}$	1	$1\frac{1}{8}$	$1\frac{3}{8}$	$1\frac{1}{2}$
E [Minimum anchor embedment (in.)]	$3\frac{3}{8}$	$4\frac{1}{2}$	$5\frac{5}{8}$	$6\frac{3}{4}$	$7\frac{7}{8}$	9	$10\frac{1}{8}$	$11\frac{1}{4}$

For **SI**: 1 inch = 25.4 mm, 1 square inch = 645.16 mm².

TABLE 9—REDUCTIONS FOR SPACING AND EDGE DISTANCES FOR REINFORCING BARS INSTALLED IN CONCRETE WITH POWER-FAST EPOXY ADHESIVE

DIMENSION / LOAD TYPE	DISTANCE FOR FULL ANCHOR CAPACITY ¹	MINIMUM DISTANCE ²	REDUCTION FACTOR ³
SPACING (s) BETWEEN ANCHORS – TENSION AND SHEAR LOADS	$s_{cr} = 18D$	$s_{min} = 9D$	$F_t = F_v = 0.50$
EDGE DISTANCE (c) – TENSION LOADS	$c_{cr} = 12D$	$c_{min} = 6D$	$F_t = 0.56$
EDGE DISTANCE (c) – SHEAR LOADS	$c_{cr} = 16D$	$c_{min} = 4D$	$F_v = 0.17$

¹The listed values are the minimum distances required to obtain the load values listed in Tables 10 and 11. D = Anchor diameter = reinforcing bar in inches. When adjacent anchors are different sizes or embedments, use largest value for D .

²The listed values are the minimum distances at which the anchor can be set, when load values are adjusted appropriately.

³Load values in the tables are multiplied by the reduction factor when anchors are installed at the minimum spacing listed. Use linear interpolation for spacing between critical and minimum distances. When a group of anchors is affected by both reduced spacing and reduced edge distance, the edge and spacing reduction factors must be combined (multiplied).

TABLE 10—ALLOWABLE TENSION LOADS FOR REINFORCING BARS INSTALLED IN NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8}

REINFORCING BAR SIZE	BIT DIAMETER (inches)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)	ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)	
					$f'_c \geq 2,000$ psi Minimum	GRADE 40	GRADE 60
No. 3	$\frac{1}{2}$	$6\frac{3}{4}$	$4\frac{1}{2}$	$3\frac{3}{8}$	2,785	2,200	2,640
No. 4	$\frac{5}{8}$	9	6	$4\frac{1}{2}$	3,610	4,000	4,800
No. 5	$\frac{3}{4}$	$11\frac{1}{4}$	$7\frac{1}{2}$	$5\frac{5}{8}$	5,030	6,200	7,440
No. 6	$\frac{7}{8}$	$13\frac{1}{2}$	9	$6\frac{3}{4}$	6,580	8,800	10,560
No. 7	1	$15\frac{3}{4}$	$10\frac{1}{2}$	$7\frac{7}{8}$	8,330	12,000	14,400
No. 8	$1\frac{1}{8}$	18	12	9	12,085	15,800	18,960
No. 9	$1\frac{3}{8}$	$20\frac{1}{4}$	$13\frac{1}{2}$	$10\frac{1}{8}$	13,930	19,800	23,760
No. 10	$1\frac{1}{2}$	$22\frac{1}{2}$	15	$11\frac{1}{4}$	18,515	24,450	29,450

For **SI**: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. When Power-Fast Epoxy Standard-Set adhesive is used with reinforcing bar, the allowable loads may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. When Power-Fast Epoxy Fast-Set adhesive is used with reinforcing bars, resistance to earthquake loads is beyond the scope of this report.

²For reinforcing bars Grade 60, the allowable steel strength is based on the tensile stresses listed in Section 1926.3.2 of the UBC, applied to the cross-section area of the rebar. For earthquake application with Grade 40 reinforcing bar, reduce the allowable steel strength using a load factor of 0.83.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 9. The reduction factors listed in Table 9 must be applied to the values listed in the bond and concrete strength column.

⁴The tabulated shear and tension values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation.

⁵Allowable loads for the Standard Set Power-Fast adhesive, based on bond strength, are calculated based on a safety factor of 4.0. When using the Fast Set Power-Fast adhesive, the bond strength must be reduced by 25 percent based on a safety factor of 5.33.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 1 for applicable load factors.

⁸Special inspection in accordance with Section 2.5 must be provided for all anchor installations.

TABLE 11—ALLOWABLE SHEAR LOADS FOR REINFORCING BARS INSTALLED IN MINIMUM 2,000 PSI NORMAL-WEIGHT CONCRETE WITH POWER-FAST EPOXY ADHESIVE^{1,2,3,4,5,6,7,8,9}

REINFORCING BAR SIZE	BIT DIAMETER (inch)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	BOND STRENGTH (pounds)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)	
						GRADE 40	GRADE 60
No. 3	$\frac{1}{2}$	$6\frac{3}{4}$	6	$3\frac{3}{8}$	2,075	1,680	1,680
No. 6	$\frac{7}{8}$	$13\frac{1}{2}$	12	$6\frac{3}{4}$	7,015	6,730	6,730

For **SI**: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength. When Power-Fast Epoxy Standard-Set adhesive is used with reinforcing bar, the allowable loads may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. When Power-Fast Epoxy Fast Set adhesive is used with reinforcing bars, resistance to earthquake loads is beyond the scope of this report.

²Allowable loads must be multiplied by 82% for short-term loading due to earthquake and wind. For earthquake and wind application with Grade 60 reinforcing bar, the allowable loads must be the lesser of the bond strength and the allowable steel strength. For earthquake application of Grade 40 reinforcing bar, reduce the allowable steel strength using a load factor of 0.83.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 9. The reduction factors listed in Table 9 must be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.0.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 1 for applicable load factors.

⁸Special inspection in accordance with Section 2.5 must be provided for all anchor installations.

⁹Concrete compressive strength is 2,000 psi.

TABLE 12—SPECIFICATIONS AND DETAILS FOR INSTALLATION OF THE CHEM-STUD THREADED ROD IN CONCRETE USING CHEM-STUD ADHESIVE

PROPERTY	ROD DIAMETER						
	$\frac{3}{8}$ inch	$\frac{1}{2}$ inch	$\frac{5}{8}$ inch	$\frac{3}{4}$ inch	$\frac{7}{8}$ inch	1 inch	$1\frac{1}{4}$ inches
A_b [Nominal area of rod (in. ²)]	0.1105	0.1963	0.3068	0.4418	0.6013	0.7854	1.2772
BD [Nominal bit diameter (in.)]	$\frac{7}{16}$	$\frac{9}{16}$	$\frac{11}{16}$	$\frac{7}{8}$	1	$1\frac{1}{8}$	$1\frac{3}{8}$
E [Minimum anchor embedment (in.)]	$3\frac{1}{2}$	$4\frac{1}{4}$	5	$6\frac{5}{8}$	7	$8\frac{1}{4}$	$10\frac{1}{2}$
T [Maximum tightening torque (ft.-lbf.)]	20	40	90	160	200	300	500

For **SI**: 1 inch = 25.4 mm, 1 square inch = 645.16 mm², 1 ft.-lbf. = 1.35 N-m.

TABLE 13—MANUFACTURER'S RECOMMENDED CURE TIME FOR CHEM-STUD ADHESIVE

BASE MATERIAL TEMPERATURE ¹ (°F)	FULL CURE TIME ²
32 to 65	1 hour
65 to 68	30 minutes
68 and above	20 minutes

For **SI**: t°C = $\frac{5}{9}(t^{\circ}\text{F} - 32)$.

¹This is the temperature as measured in the base material.

²The full curing time is the minimum time required for the adhesive to achieve its ultimate load capacities.

TABLE 14—REDUCTIONS FOR SPACING AND EDGE DISTANCES FOR CHEM-STUD THREADED ROD INSTALLED IN CONCRETE WITH CHEM-STUD ADHESIVE

DIMENSION / LOAD TYPE	DISTANCE FOR FULL ANCHOR CAPACITY ¹	MINIMUM DISTANCE ²	REDUCTION FACTOR ³
SPACING (s) BETWEEN ANCHORS – TENSION AND SHEAR LOADS	$s_{cr} = 16D$	$s_{min} = 8D$	$F_t = F_v = 0.64$
EDGE DISTANCE (c) – TENSION LOADS	$c_{cr} = 10D$	$c_{min} = 4D$	$F_t = 0.64$
EDGE DISTANCE (c) – SHEAR LOADS	$c_{cr} = 12D$	$c_{min} = 4D$	$F_v = 0.22$

¹The listed values are the minimum distances required to obtain the load values listed in Tables 14 and 15. D = Anchor diameter. When adjacent anchors are different sizes or embedments, use largest value for D .

²The listed values are the minimum distances at which the anchor can be set, when load values are adjusted with the reduction factor noted in this table.

³Load values in the tables are multiplied by the reduction factor with the reduction factor noted in this table. Linear interpolation is allowed for edge distances and spacing that fall between critical and minimum distances. When a group of anchors is affected by both reduced spacing and reduced edge distance, the edge and spacing reduction factors must be combined (multiplied).

TABLE 15—ALLOWABLE TENSION LOADS FOR THE CHEM-STUD THREADED ROD INSTALLED IN NORMAL-WEIGHT CONCRETE WITH CHEM-STUD ADHESIVE^{1,2,3,4,5,6,7,8}

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s_{cr} (inches)	EDGE DISTANCE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE TENSION LOAD BASED ON BOND OR CONCRETE CAPACITY (pounds)				ALLOWABLE TENSION LOAD BASED ON STEEL STRENGTH (pounds)			
					$f'_c \approx 2,000$ psi	$f'_c \approx 3,000$ psi	$f'_c \approx 4,000$ psi	$f'_c \approx 5,000$ psi	ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
$3/8$	$7/16$	6	$3 3/4$	$3 1/2$	1,575	1,885	2,190	2,805	2,115	2,185	4,555	3,630
$1/2$	$9/16$	8	5	$4 1/4$	2,510	2,950	3,390	4,270	3,755	3,885	8,095	6,470
$5/8$	$1 1/16$	10	$6 1/4$	5	3,460	4,060	4,655	5,850	5,870	6,075	12,665	10,130
$3/4$	$7/8$	12	$7 1/2$	$6 5/8$	5,575	6,355	7,135	8,695	8,455	8,750	18,225	12,400
$7/8$	1	14	$8 3/4$	7	5,870	7,005	8,140	10,410	11,510	11,905	24,805	16,860
1	$1 1/8$	16	10	$8 1/4$	8,090	9,885	11,675	15,260	15,035	15,550	32,400	22,020
$1 1/4$	$1 1/2$	20	12	$10 1/2$	14,595	16,225	17,850	20,560	23,485	24,295	50,615	34,420

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength.

²For Earthquake and wind application with A 193 Grade B7 threaded rod, the lesser of the tabulated bond strength and the allowable steel strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. For earthquake and wind application with SS304/316 threaded rod, reduce the allowable steel strength using a load factor of 0.92. The lesser of this reduced steel strength and tabulated bond strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. For earthquake application with A 36 and A 307 threaded rod, reduce the allowable steel strength using a load factor of 0.48. The lesser of this reduced steel strength and tabulated bond strength may be increased for duration of load in accordance with Section 1612.3.3 of the UBC. For wind application with A 36 and A 307 threaded rod, the allowable load must be the lesser of the tabulated bond strength and the allowable steel strength.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 14. The reduction factors listed in Table 14 must be applied to the values listed in the bond and concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.0.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷Special inspection in accordance with Section 2.5 must be provided for all anchor installations.

⁸Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 2 for applicable load factors.

TABLE 16—ALLOWABLE SHEAR LOADS FOR THE CHEM-STUD THREADED ROD INSTALLED IN MINIMUM 2,000-PSI NORMAL-WEIGHT CONCRETE WITH CHEM-STUD ADHESIVE^{1,2,3,4,5,6,7,8}

ANCHOR DIAMETER (inches)	BIT DIAMETER (inches)	SPACING, s_{cr} (inches)	EDGE, c_{cr} (inches)	MINIMUM EMBED. (inches)	ALLOWABLE SHEAR LOAD BASED ON STEEL STRENGTH (pounds)			
					ASTM A 36	ASTM A 307	ASTM A 193 B7	304 / 316 SS
$3/8$	$7/16$	6	$4 1/2$	$3 1/2$	1,090	1,125	2,345	1,870
$1/2$	$9/16$	8	6	$4 1/4$	1,940	2,000	4,170	3,330
$5/8$	$1 1/16$	10	$7 1/2$	5	3,025	3,130	6,520	5,210
$3/4$	$7/8$	12	9	$6 5/8$	4,355	4,505	9,390	6,390
$7/8$	1	14	$10 1/2$	7	5,930	6,135	12,780	8,680
1	$1 1/8$	16	12	$8 1/4$	7,745	8,010	16,690	11,340
$1 1/4$	$1 1/2$	20	15	$10 1/2$	12,100	12,515	26,075	17,730

For SI: 1 inch = 25.4 mm, 1 psi = 6.89 kPa, 1 lbf = 4.48 N.

¹Allowable load must be the lesser of the bond or steel strength.

²For short-term loading due to earthquakes or wind, allowable shear loads for A 36 and A 307 threaded rods may be increased by $33 1/3\%$. A 36 and A 307 values must be used instead of listed values for higher strength steels for these loading conditions.

³The tabulated values are for anchors installed with edge distance (c_{cr}) and spacing (s_{cr}) in accordance with Table 14. The reduction factors listed in this table must be applied to the values listed in the bond or concrete strength column.

⁴The tabulated load values are for anchors installed in normal-weight concrete that has reached the designated ultimate compressive strength at the time of installation. Linear interpolation may be used for concrete strengths between those listed.

⁵Allowable loads based on bond strength are calculated based on a safety factor of 4.0.

⁶The minimum member or slab thickness must be greater than 1.5 times the anchor embedment depth.

⁷Special inspection in accordance with Section 2.5 must be provided for all anchor installation.

⁸Adhesive anchors experience a reduction in tensile and shear capacity with increased ambient temperatures. See Figure 2 for applicable load factors.

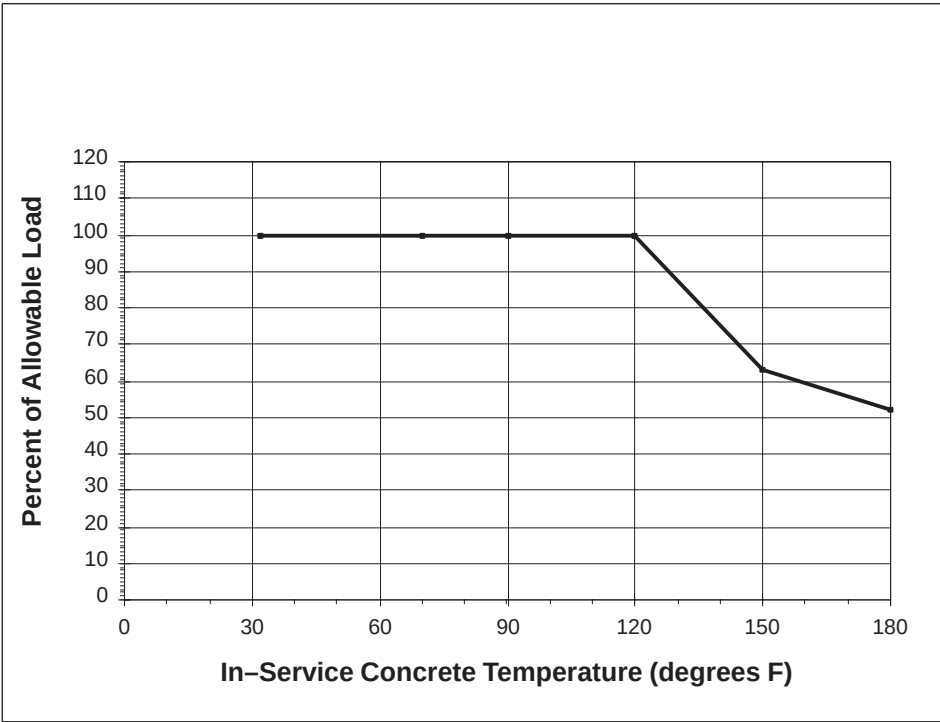


FIGURE 1—ALLOWABLE LOAD BOND STRENGTH REDUCTION
BASED ON IN-SERVICE TEMPERATURE FOR THE POWER-FAST EPOXY ADHESIVE

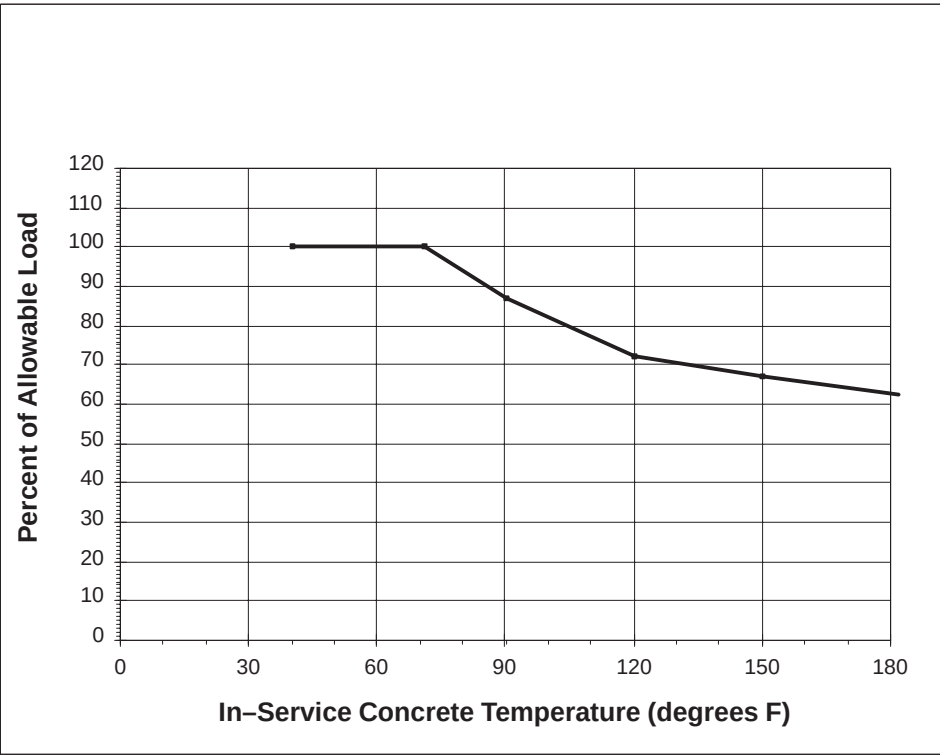


FIGURE 2—ALLOWABLE LOAD BOND STRENGTH REDUCTION
BASED ON IN-SERVICE TEMPERATURE FOR THE CHEM-STUD ADHESIVE